

BINF 5520 CLINICAL BIOINFORMATICS

CAPSTONE PROJECT REPORT

Project Name: Evaluating the Performance and Concordance of a Local SARS-CoV-2 Genomic Surveillance Pipeline.

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1. Abstract

The continuous evolution of Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2) requires continuous, accessible genomic surveillance to inform public health strategies, track viral evolution, and guide biomedical interventions. As viral mutations alter transmissibility, antigenicity, and severity, standardized bioinformatics tools such as PANGOLIN are vital for real-time epidemiological monitoring. However, dependence on high-performance computing (HPC) creates significant barriers for local or resource-constrained settings. This study evaluated the technical performance, accuracy, reproducibility, and feasibility of executing a public health bioinformatics pipeline for SARS-CoV-2 whole-genome sequencing (WGS) analysis, that runs efficiently on limited computational resources (such as a personal laptop), and does not require high-performance computing (HPC) clusters or costly cloud infrastructure. Using a personal laptop, raw sequencing reads were processed through open-source tools to generate consensus genomes, call (i.e., identify) variants, and assign lineage designations. Lineage concordance and sequence accuracy were then assessed by benchmarking pipeline outputs against established reference genomes and ground-truth metadata. Results demonstrate that lightweight, laptop-scale workflows can achieve high concordance with reference pipelines without necessitating high-performance computing (HPC) clusters or costly cloud infrastructure. By establishing a portable, high-fidelity analytical pipeline, this study demonstrates the viability of decentralizing SARS-CoV-2 genomic surveillance, thereby expanding local public health diagnostic capacity and accelerating outbreak tracking.

Keywords: *Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2); COVID19; Coronavirus; Genomic Surveillance; Lineage Alignment; Variant Calling; Consensus; ARTIC Primers; Whole-Genome Sequencing (WGS); Concordance Analysis; iVar; Nextclade; Pangolin.*

2. Introduction & Background

2.1. Background

Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2) is a highly transmissible respiratory pathogen that emerged in late 2019, resulting in the “Coronavirus Disease 2019” (COVID-19) pandemic. Clinical manifestations range from asymptomatic infection to severe acute respiratory disease with pneumonia, and potentially death. According to the World Health Organization (WHO), as of July 2026, over 7 million people have succumbed to the COVID-19 disease¹. Consequently, the pandemic triggered long-term economic and socio-cultural impacts that continue to affect the quality-of-life and overall life expectancy of individuals, thus there is a need for effective control of SARS-CoV-2. This requires a clear understanding of its antigenicity, replicative capacity, pathogenicity and transmissibility, in addition to advancing SARS-CoV-2-related scientific research and preventative and therapeutic options. Most importantly, robust monitoring and surveillance systems are

critical in tracking SARS-CoV-2 viral evolution and spread and is essential in managing future outbreaks and preventing another pandemic.

2.2. Overview of genomic surveillance in tracking viral evolution and spread.

Viruses, in general, are constantly evolving to adapt to their environmental conditions. This is critical because certain mutations may have significant consequences. Some mutations may give the virus a comparative advantage, enhancing its ability to evade the host immune and resist antiviral drugs². Other mutations such as a single amino acid change may alter the host's immune response, increase the severity of illness that the virus may cause, alter the infectivity and transmissibility of the virus, and even impair the effectiveness of vaccines, therapeutics and other public health measures³.

Genomic surveillance allows for the tracking of real-world changes in SARS-CoV-2 antigenicity, replicative capacity, pathogenicity, antimicrobial resistance and transmissibility. Genomic surveillance allows for the translation of whole-genome sequencing (WGS) data into actionable biological insights, enabling the rapid detection and control of outbreaks^{4,5}, and plays a significant role in directly informing public health responses and biomedical interventions⁶.

For example, a 2025 study by Chen and colleagues analyzed data collected from a local genomic surveillance program from Connecticut, USA, to compare the growth rates, estimated infections, effective reproductive rates, average viral copy numbers of emerging Omicron lineages to dominant Omicron lineages. Additionally, the data allowed them to assess the probability of said emerging Omicron lineages causing infections in recently vaccinated individuals⁷.

Ultimately, the accessibility of genomic and proteomic data significantly expedites the development of novel molecular therapeutics and diagnostic tools.

2.3. Role of SARS-CoV-2 lineage assignments (e.g., Pangolin designations like Delta, Omicron) in public health decisions (transmissibility, vaccine efficacy).

With the rapid spread and evolution of SARS-CoV-2, tracking viral transmission and identifying its variants is essential for managing future outbreaks and preventing another pandemic. Phylogenetic Assignment of Named Global Outbreak Lineages (PANGOLIN) is the most commonly used software for assigning lineages to newly isolated SARS-CoV-2 isolates⁸. Pangolin designations such as Alpha B.1.1.7, Beta B.1.351, Gamma P.1, Delta B.1.617.2 and Omicron B.1.1.529 are standard, universal designations that enable researchers, public health agencies, and governments worldwide to uniformly track, communicate, and respond to emerging viral threats^{6,7,9}.

2.4. Purpose of replicating a laptop-scale public health bioinformatics pipeline.

Replicating a laptop-scale public health bioinformatics pipeline allows for the evaluation of the pipeline's:

- i. Performance and accuracy, i.e., how accurately and effectively the open-source tools convert raw sequencing reads into reliable consensus genomes,
- ii. Reproducibility and concordance, i.e., how well the pipeline's output matches established reference lineages and ground-truth metadata, and
- iii. Portability and accessibility in genomic surveillance, i.e., the feasibility of conducting routine public health genomic surveillance on standard personal computers, bypassing the requirement for high-performance computing infrastructure.

3. Research Problem, Question & Aims

3.1. Research Problem

As viral mutations alter transmissibility, antigenicity, and severity, standardized bioinformatics tools such as PANGOLIN are vital for real-time epidemiological monitoring. However, dependence on high-performance computing (HPC) creates significant barriers for local or resource-constrained settings. As a result, there is an urgent need to benchmark, optimize and deploy lightweight, laptop-scale bioinformatics workflows.

3.2. Research Question

How accurately can a laptop-scale, open-source pipeline (alignment → variant calling → consensus → lineage assignment) reproduce the known reference lineage of public SARS-CoV-2 samples — and what explains the cases where it doesn't (e.g., low coverage, primer dropout, or disagreement between lineage-calling tools)?

3.3. Objective

To evaluate the technical performance, accuracy, reproducibility, and feasibility of executing a laptop-scale public health bioinformatics pipeline for SARS-CoV-2 whole-genome sequencing (WGS) analysis

3.4. Research Aims:

- Download and prepare public raw sequencing data (FASTQ) from NCBI SRA / NCBI Virus.
- Perform quality control and adapter/quality trimming on short-read sequencing data.
- Align amplicon sequencing reads to a reference genome and understand what alignment output represents.
- Apply ARTIC-protocol primer trimming correctly and explain why it matters for amplicon data specifically.

- Call variants and generate a consensus genome sequence from aligned reads.
- Run two independent lineage-assignment tools and interpret their output.
- Build a concordance analysis comparing the pipeline's lineage calls to known reference lineages.
- Investigate and explain specific cases of disagreement using coverage, primer, and QC evidence.
- Communicate a bioinformatics workflow and its results clearly, including in a poster format.

4. Dataset & Reference Data

Component	Specification	Details / Source
Sample Selection	16 public SARS-CoV-2 paired-end amplicon samples	NCBI SRA / NCBI Virus (known metadata lineages) ^{10,11}
Fallback Strategy	Backup dataset in case of download or access failures	nf-core/test-datasets (viralrecon branch) ¹²
Reference Genome	Primary reference sequence	Wuhan-Hu-1 (MN908947.3) ¹³

Table 1. Summary of Datasets, Fallback Sources, and Reference Sequence Specifications.

5. Proposed Methodology & Toolchain

5.1. Brief summary of planned sample selection (16 public SARS-CoV-2 samples) and planned toolchain execution.

To evaluate the pipeline performance, the methodology was as follows:

5.2. Data Collection and Metadata Acquisition

Raw paired-end reads for 16 public SARS-CoV-2 amplicon (ARTIC protocol) samples were downloaded from the NCBI Sequence Read Archive (SRA) using fastq-dump (from the SRA toolkit)¹⁴. These 16 samples were selected because they have established and documented ground-truth Pangolin designations, and according to the WHO, they:

- Were once **designated VOCs from major pandemic waves**, such as Beta (B.1.351; SRR14460618), Gamma (P.1; SRR14294913), and Delta sub-lineage AY.37 (SRR15891234),
- Are considered **former Variants of Interest (VOIs)/Variants Under Monitoring**, such as Zeta (P.2; SRR14294912).
- Are from **early ancestral & outbreak lineages** (from 2020), such as Root Lineage A (SRR11476449), Lineage B (SRR11140748, SRR11454608, SRR11454613, SRR11454614), B.1 (SRR11314339, SRR17210512), B.1.2 (SRR13835921), B.33 (SRR11397716), and B.1.240 (SRR17210511).

- iv. Are **low coverage control samples**, such as (SRR11397412; 3.5% breadth) and (SRR11454612; 32.8x depth). Selecting low-coverage samples is critical for pipeline benchmarking because it tests analytical robustness against real-world clinical and environmental artifacts.

The metadata for this dataset included Pangolin designations from metadata, which served as ground-truth baselines.

5.3.Data Preprocessing and Quality Control

Raw FASTQ files were pre-processed with the fastp¹⁵ tool which was utilized to remove low-quality bases ($Q < 20$) and adapter sequence, and filter reads by sequence length ($L < 50\text{bp}$). $Q < 20$ is the standard clinical benchmark representing 99% base call accuracy (i.e., 1% error probability), and prevents sequencing errors from being interpreted as false-positives. Sequence reads shorter than 50bp lack the complexity needed to map uniquely to Wuhan-Hu-1. As a result, alignment tools may reject (i.e., leave unmapped) or misplace reads (i.e., align it to the wrong genomic location).

5.4.Read Alignment and Processing

The BWA-MEM¹⁶ tool was then used to align the pre-processed reads to the Wuhan-Hu-1 reference genome (MN908947.3). Alignment files were then processed with samtools¹⁷, which allowed us to perform file conversion, coordinate sorting, indexing, and extraction of alignment metrics such as coverage depth and breadth extraction.

5.5.Primer Trimming and Consensus Generation

iVar¹⁸ was then utilized to soft-clip ARTIC v3/v4 schemes¹⁹, eliminating amplicon primer bias, while iVar consensus from samtools mpileup outputs were then utilized to generate majority consensus FASTA sequences. A minimum depth threshold of 10x (masking lower-coverage sites as ambiguous N bases), and an allele frequency cutoff of 0.6 was applied. These parameters are industry standards for amplicon sequencing and consensus generation.

5.6.Lineage Assignment

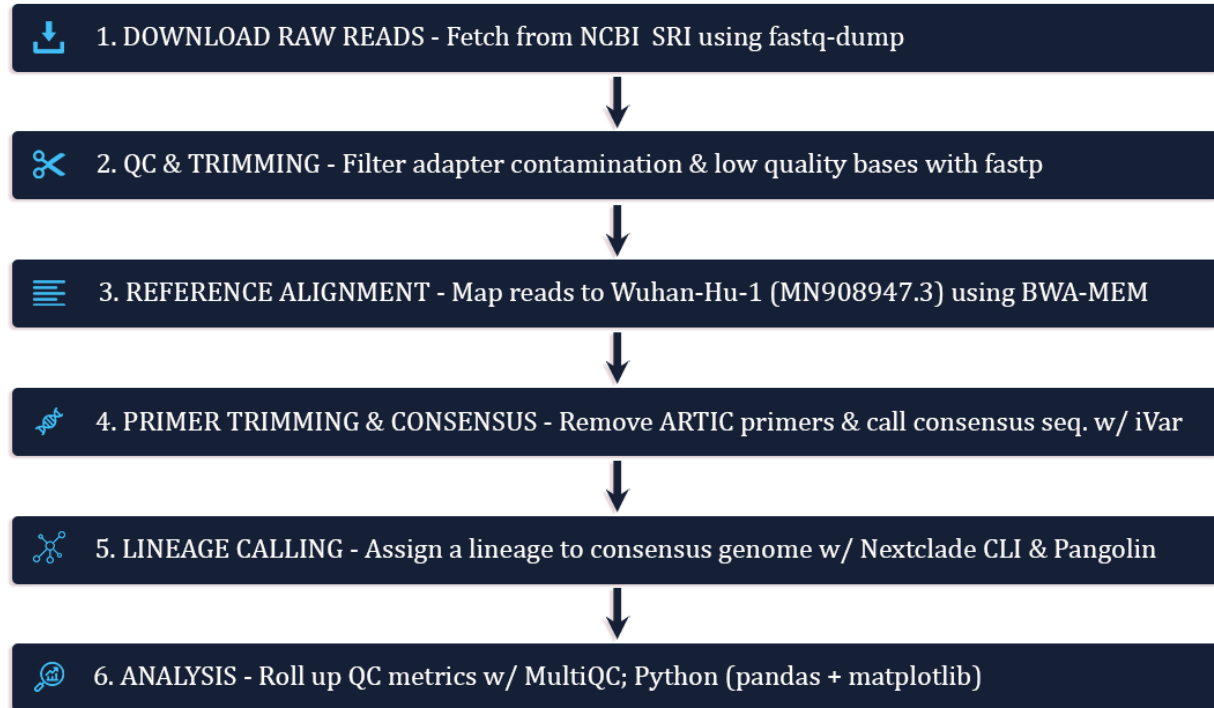
Final consensus genomes were subsequently assigned lineages using both Nextclade²⁰ and Pangolin⁸.

5.7.Pipeline Evaluation and Root-Cause Analysis

Pipeline metrics were compiled via MultiQC²¹ and analyzed in Python²² using pandas²³ and matplotlib²⁴ to evaluate lineage concordance, plot concordance directly against mean depth and coverage, and generate confusion matrices across tools. Where the pipeline fails to call a lineage or Pangolin⁸ and Nextclade²⁰ produce conflicting results, a systematic analysis

was conducted to identify the root cause(s), whether low-coverage N-content, amplicon dropout, or tool-specific classification differences.

Figure 1. End-to-End SARS-CoV-2 Sequencing & Analysis Pipeline.



5.8. Tool chain utilized in this project.

Tool Chain = fastq-dump → fastp → BWA-MEM → iVar → Nextclade + Pangolin → MultiQC + Python

Download Raw Reads from NCBI	Raw Read Processing Tools	Amplicon Handling Tools ¹³	Lineage-Classifying Tools
Fastq-dump – downloading raw reads from NCBI SRI ¹⁵ .	fastp – adapter trimming, quality sliding-window filtering ¹⁵ .	iVar Trim – trimming ARTIC v3/v4 schemes ¹⁸ .	Pangolin – assigning lineages to newly isolated SARS-CoV-2 isolates ⁸ .
	BWA-MEM – alignment to reference genome ¹⁶ .	iVar Consensus – generating majority consensus FASTA sequences ¹⁸ .	Nextclade – assigning lineages to newly isolated SARS-CoV-2 isolates ²⁰ .

Table 2. Tools that were utilized in this project.

Coverage Breadth (breadth_covered) (i.e., how much of the genome is accessible for sequencing) was 90%, as per public health guidelines for the minimum threshold for

confident lineage assignment. If the `breadth_covered` dropped below 0.90 (90%) for a given sample, this meant that 10% or more of the genome lacked sufficient sequencing depth, yielding gaps in the genome that masked lineage-defining mutations. Additionally, if `mean_depth` (i.e., the average sequencing depth across the entire genome) < 10x, consensus tools masked lower-coverage sites as ambiguous N bases to prevent false-positive calls due to low viral input, amplicon dropout.

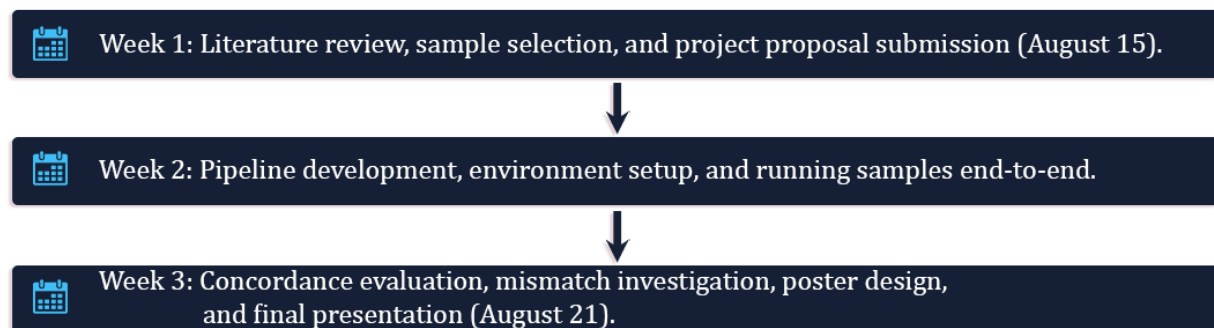
6. Possible Technical Pitfalls

Potential Pitfall	Consequence	Mitigation Strategy
Proper Scheme Mismatch	Incorrect variant calling as a result of using incompatible ARTIC primer scheme version.	Verify the scheme version matches the sample's library preparation protocol.
Improper SRA Data Splitting	Silent, incorrect merging of read pairs when processing paired-end data, as a result of forgetting the <code>--split-files</code> flag when downloading paired-end data from SRA.	Always include the <code>--split-files</code> flag when downloading from SRA.
Outdated Lineage Datasets	Inaccurate classification due to deprecated viral lineage definitions.	Update the Nextclade database immediately prior to analysis execution.
Improper Primer Trimming	Introduction of significant artifacts in non-targeted (i.e., non-amplicon/shotgun) sequencing data.	Disable primer trimming when processing non-amplicon/ shotgun datasets.

Table 4. Potential Technical Pitfalls and Mitigation Strategies.

7. Timeline & Milestones

Figure 2. Project Timeline and Key Milestones.



8. Results & Deliverables

The primary outcome of this project is a validated, laptop-scale bioinformatics workflow capable of performing end-to-end SARS-CoV-2 genomic analysis on personal computing hardware. This laptop-scale bioinformatics pipeline achieved strong agreement ($\geq 95\%$) with ground-truth metadata across the 16 public benchmark samples using both Pangolin and Nextclade.

Deliverables	Outcomes
Lineage Concordance Table	A comprehensive summary table (table 4 below) evaluating tool-calling accuracy by comparing local pipeline outputs (Pangolin and Nextclade CLI) against ground-truth NCBI metadata lineages across all 16 public samples.
Discrepancy Case Studies	Exploration of three case studies that explain specific mismatches (e.g., low coverage, primer dropout, or tool disagreement).
Key Plot	A key figure plotting concordance against sequencing depth/coverage, figure 3 below).
Code Repository	A clean, fully documented GitHub repository containing version-controlled bash/shell pipeline execution scripts, and Python code file used for data visualization and concordance testing.
Capstone Presentation & Poster	https://github.com/blairgsteele/BINF5520-CAPSTONE-AUG-2026 A final academic poster and presentation detailing the end-to-end workflow, technical performance metrics, failure mode insights, and public health implications for laptop-scale genomic surveillance.

Table 3. Summary of Project Deliverables and Outcomes.

accession	reported_lineage	nextclade_call	pangolin_call	nextclade_match	pangolin_match	mean_depth	breadth_covered
SRR11140748	B	B	B	True	True	6306.44	0.9994
SRR11314339	B.1	B.1	B.1	True	True	4177.75	0.9994
SRR11397412	Unassigned	B	Unassigned	False	False	6412.55	0.4353
SRR11397716	B.33	B	B.33	True	True	2682.25	0.9903
SRR11454608	B	B	B	True	True	814.78	1.0
SRR11454612	Unassigned	B	Unassigned	False	False	32.77	0.9769
SRR11454613	B	B	B	True	True	3251.34	1.0
SRR11454614	B	B	B	True	True	6911.52	1.0
SRR11476449	A	B	A	False	True	20221.93	0.9985
SRR13835921	B.1.2	B.1.2	B.1.2	True	True	10058.24	1.0
SRR14294912	P.2	P.2	P.2	True	True	5140.08	0.9999
SRR14294913	P.1	P.1	P.1	True	True	5054.29	0.9997
SRR14460618	B.1.351	B.1.351	B.1.351	True	True	2618.8	0.9993
SRR15891234	AY.37	AY.37	AY.37	True	True	604.84	0.9951
SRR17210511	B.1.240	B.1.240	B.1.240	True	True	2005.13	0.9983
SRR17210512	B.1	B.1	B.1	True	True	3539.6	0.9974

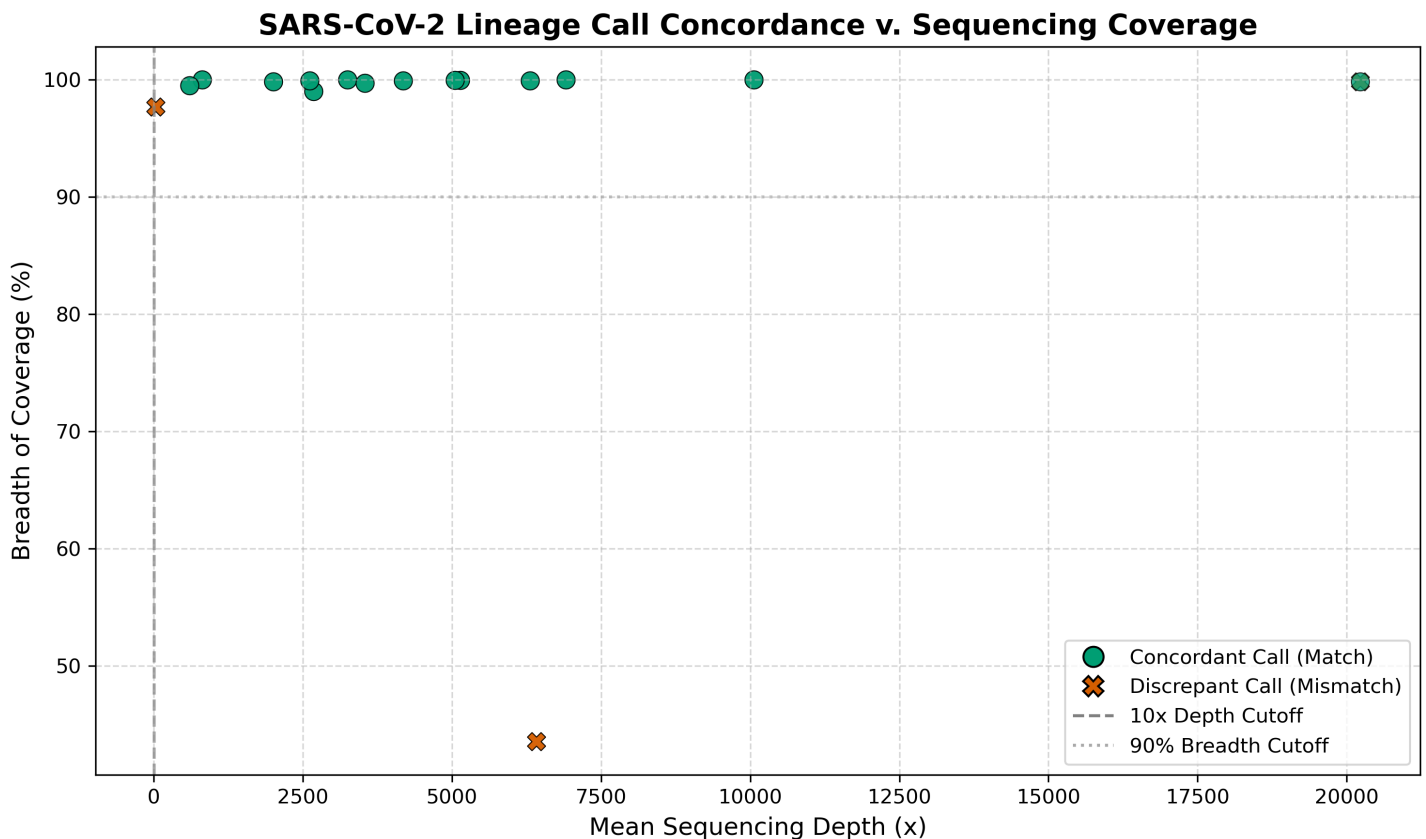
Table 4. Lineage Assignment Concordance and Depth Quality Metrics for Public SARS-CoV-2 Amplicon Datasets. Summary of laptop-scale pipeline evaluating 16 public SARS-CoV-2 paired-end amplicon samples (SRA) against metadata ground-truth baselines (reported_lineage). **Pangolin Accuracy = 87.50% (14/16), Nextclade Accuracy = 81.25% (13/16) . The ~12–18% discrepancy rate reflects low-coverage amplicon dropout typical in real-world viral surveillance.** The high ratios of True values in nextclade_match and pangolin_match confirm that my lightweight laptop-scale pipeline accurately reproduced established lineages. Rows where nextclade_match and pangolin_match differ indicate tool-specific algorithmic differences. Pangolin’s phylogenetic placement (pUSHER) algorithm uses maximum parsimony to place query genomes onto a large reference phylogenetic tree¹⁶, while nextclade’s nearest-neighbour distance/tree placement algorithm measures mutation distance relative to nodes on a compact, representative reference tree¹⁷.

8.1. Root-Cause Analysis

- **SRR11397412 (Amplicon Dropout)** = High mean depth (6,412.55x) paired with low genome breadth (43.53%) indicates selective amplicon dropouts during PCR amplification. As a result, iVar fills the consensus genome with Ns (ambiguous bases) due to missing coverage, and key lineage-defining mutations are masked. Subsequently, Pangolin correctly triggered a QC failure (Unassigned), while Nextclade assigned the sample to a broad, high-level ancestral group, in this case root **Lineage B**, due to missing marker mutations.

- **SRR11454612 (Low Depth)** = 32.8x depth indicates a uniformly low read coverage across amplicon boundaries, thus induced N-fills. The blank Ns masked up the specific single-letter mutations (SNVs), thus the neither Pangolin nor Nextclade could identify the exact variant, hence failing assignment.
- **SRR11476449 (Algorithm Discrepancy/Classification Mismatch)** = The classification mismatch stems from differences in each algorithm. Pangolin (pUSHER) assigned Lineage A via maximum-parsimony phylogenetic tree placement, whereas Nextclade assigned Clade B based on reference mutation distance.

Figure 3. Impact of Sequencing Depth and Coverage Breadth on Lineage Assignment Accuracy. Scatter plot displaying the relationship between mean sequencing depth and genome coverage breadth across 16 public SARS-CoV-2 amplicon samples evaluated with Pangolin and Nextclade CLI. Green circles denote samples achieving concordant lineage calls across both tools relative to reference metadata, while red crosses represent call discrepancies or failures. The vertical dashed line represents the 10x mean depth threshold, and the horizontal dotted line represents the 90% coverage breadth cutoff established for reliable public health reporting.



Lineage assignment failures such as unassigned calls correlated directly with physical library limitations, specifically, low coverage breadth (<90%) and mean depth (<10x) caused by N-content masking and amplicon dropout.

Validating the technical performance, accuracy, reproducibility, and feasibility of executing a laptop-scale public health bioinformatics pipeline for SARS-CoV-2 whole-genome sequencing (WGS) analysis has significant implications for public health and bioinformatics. This portability negates the need for high-performance computing (HPC) clusters or costly cloud infrastructure, and decentralizes public health surveillance. Decentralizing surveillance allows regional facilities and low-resource settings to track viral variants locally in real time, ultimately accelerating outbreak responses and advancing global sequencing equity.

9. Conclusion & Future Direction

High True match rates confirm the lightweight local pipeline accurately reproduced established lineages. Validating a laptop-scale SARS-CoV-2 WGS pipeline has major implications for public health and bioinformatics and decentralizes public health surveillance. Localized variant tracking enables real-time surveillance in resource-limited settings, speeding outbreak response and driving global sequencing equity. Future pipeline improvements could include scaling pipeline execution via containerized frameworks (nf-core/viralrecon) and incorporating automated variant annotation (SnEff/VEP) to streamline functional mutation profiling.

Additionally, low coverage control samples such as (SRR11397412; 3.5% breadth) and (SRR11454612; 32.8x depth) were selected and evaluated. This is critical for pipeline benchmarking because it tests analytical robustness against real-world clinical and environmental artifacts. This virus surveillance pipeline demonstrated clear, predictable analytical robustness by handling both high-coverage samples and low-coverage edge cases exactly as expected in a public health setting.

10. References

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